

- 1 Root hair cells are specialised plant cells located in the outer layer of young roots of plants. Root hair cells have an essential role in the uptake of water and dissolved mineral ions from the soil solution.

The transport of water across the root to reach the central xylem tissue can occur by the symplast pathway or apoplast pathway.

- (a) Fig. 1.1 is a diagram of a photomicrograph of a root hair cell.

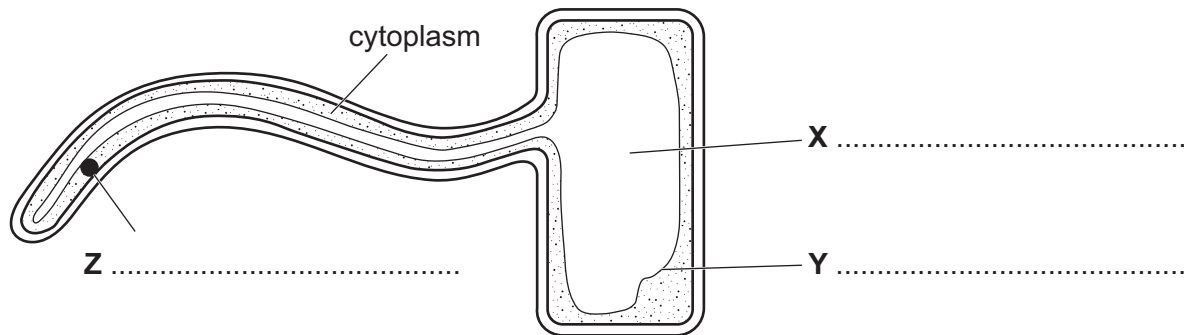


Fig. 1.1

On Fig. 1.1, name cell structures X, Y and Z.

[3]

- (b) In the symplast pathway, water passes through the cells of the different tissues in the root before entering the xylem vessels of xylem tissue.

Name the tissues of the root, in the correct sequence, through which water passes in the symplast pathway.

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- (c) In the apoplast pathway, water passes along the cell walls of adjacent plant cells and through the intercellular spaces. This is more efficient than the symplast pathway.

Explain the structural features of plant cell walls that make the apoplast pathway an efficient pathway for the transport of water.

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(d) Root hair cells also synthesise and secrete substances into the soil.

Electron microscopy of the structure of root hairs has identified endoplasmic reticulum (ER), a number of small Golgi bodies, and numerous vesicles.

- Root hair cells of the barley plant secrete enzymes known as acid phosphatases, which catalyse the release of inorganic phosphate ions from organic phosphates in the soil.
- Root hair cells of the sorghum plant secrete a hydrophobic, lipid compound known as sorgoleone, which slows down the growth of neighbouring plants.

(i) Explain why the proportion of rough ER to smooth ER may be different in the root hair cells of barley plants compared with the root hair cells of sorghum plants.

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(ii) Student **X** stated that acid phosphatases and sorgoleone could be transported out of root hair cells using the **same** process.

Student **Y** stated that acid phosphatases and sorgoleone are transported out of root hair cells using **different** processes.

Suggest the reasons given by student **X** and by student **Y** to support their statements.

student **X**

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student **Y**

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[Total: 14]